Fiscal Year:	FY 2007	**Task Last Updated:**	FY 02/26/2008
PI Name:	Brown, Christopher Ph.D.		
Project Title:	A Genomic Analysis of the Effects of Space Flight on Transgenic Plants with Compromised Signaling Pathways		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27695-7006	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2004 Space Life Sciences 04-OBPR-01: ILSRA 2004
Start Date:	01/01/2005	**End Date:**	05/15/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date corrected to 5/15/2007 per PI (5/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Boss, Wendy (North Carolina State University) Winter Sederoff, Heike (North Carolina State University) Perera, Imara (North Carolina State University)		
Grant/Contract No.:	ILSRA-04-0071		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Little is known about global gene expression patterns in organisms exposed to the microgravity environment of space. We propose to use comparative DNA microarray technology to reveal changes in the transcript levels (mRNA) that are specifically mediated by the inositol 1,4,5-trisphosphate (InsP3) signaling pathway during spaceflight. Our hypothesis is that InsP3 is an important second messenger for sensing environmental stimuli (including microgravity) and that dampening the InsP3 signal will attenuate responses by InsP3-mediated signal transduction cascades and thereby reveal InsP3-regulated processes.This proposal focuses on a model organism (Arabidopsis thaliana) that has been genetically engineered so that InsP3 signaling is dampened, therefore providing a system for investigating the roles of InsP3-mediated gene expression. The transgenic plants express the human gene that encodes type I inositol polyphosphate 5-phosphatase, an enzyme that specifically hydrolyzes InsP3 and dampens the InsP3 signal. Our specific aim is to identify and characterize changes in transcript levels that are specifically mediated by InsP3 during spaceflight. We will compare transcript profiles of transgenic plants that are grown in space and on the ground with wild type plants grown in space and on the ground. This differential analysis (wild type v. transgenic in space; wild type v. transgenic on the ground; space v. ground) will reveal critical spaceflight-induced, InsP3-mediated transcriptional changes and will enable us to subtract out transcriptional changes caused by non-gravitational factors such as temperature, gas composition, relative humidity, or mechanical stress. Because this is a ubiquitous pathway, understanding the fundamental aspects of spaceflight-induced, InsP3-mediated signaling will be applicable to eukaryotic systems.We are strongly committed to increased understanding of Fundamental Space Biology research through formal and informal education opportunities. As part of the effort, we intend to support one NASA Fundamental Space Biology/ Kenan Fellow to develop curricular resources in accordance with national and state standards.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research of the awarded "closeout"-proposal determines the stability of RNA in Arabidopsis root tips under different conditions for sample return and processing on microarrays. This might well have impacts on the success of future research projects involved in the genomics of living organisms during space flight or for sample return from remote areas on Earth.
Task Progress:	Until full molecular analysis is possible in an “on-site” laboratory, samples resulting from experiments on the ISS will have to be stored on the station and during return flight before they can be analyzed. Commonly in ground-based experiments these samples can be either chemically or physically preserved and stored to prevent metabolic changes or degradation of the tissue/molecules during storage. These options do not always exist during space flight and return. Several chemicals are harmful to the crew, and low or ultralow temperatures are energy consuming and their availability is therefore limited. Long-term preservation of RNA in fixed biological samples will be critical to many experiments that eventually might be conducted in space. In our original proposal, we had planned to use RNAlater, which is approved for use in space and has been used successfully by other spaceflight investigators (Paul et al., 2005) to preserve the samples until they can be extracted. We have used RNAlater for ground-based experiments to preserve root apex RNA prior to large-scale RNA isolations for microarray analysis (Kimbrough et al., 2004). We began initial studies on a variety of RNAlater formulations and storage conditions to determines the most effective scenario for an ISS experiment. This experiment was specific for transgenic Arabidopsis thaliana lines expressing the human Inositolpolyphosphate-5 Phosphatase gene which have been shown to have altered responses to gravity, light, and drought stress (Perera et al., 2006; Salinas-Mondragon et al., 2005). For each experiment, Seedlings from wild type, empty vector control, 2 independent transgenic lines T206, T2-8) were grown on vertical plates (50 seedlings per plate, 3 biological replicates per treatment) in the dark for 7 days and harvested into RNA later at room temperature to immediately stop metabolism. Stability of RNA in planta will be tested in monthly intervals for 6 month. To test changes in quality and quantity, total RNA will be extracted, mRNA purified. Quantity and quality of mRNA was analyzed by spectrophotometry, gel-electrophoresis, and in the the bioanalyzer (electropherogram). RESULTS: In a first experiment we tested the stability of the harvested material incubated at either +4 degrees C or -20 degrees C in one of the following reagents: a) RNAlater (Ambion), b) RNAlater + 1% Lithium chloride c) RNAlater + 1% Aurintricatocylic acid e) Ethanol Samples were taken in monthly intervals for 6-12 month. Total RNA was extracted. Quality and quantities were monitored using spectrometry. For some key samples Electropherograms were generated and analyzed. No significant differences in quantity or quality of extracted and purified mRNA were observed. In a second experiment with direct relevance to our original proposal we tested the stability of mRNA in Arabidopsis seedlings from wild type and transgenic plants stored in RNAlater at room temperature. Twenty seeds from each line were grown in small containers (plastic cuvettes) in Agar-medium containing MS salts for 7 days. They were fully submerged in RNAlater and stored at room temperature (23 C) for 0 (control), 2, 4, or 6 days. RNA quantities and qualities as monitored by spectrophotometry and electropherometry. While the spectrophotometric data showed no significant difference in quality and quantity of purified mRNA extracted after 2, 4, or 6 days of incubation at room temperature. However in the electropherograms, non-specific degradation of mRNA was detected in all samples (wt and transgenic lines T26, T28, and C9) after 4 days and 6 days of room temperature. Stability of mRNA in the transformation control plants C9 showed consistently less degradation in the electropherograms and gel electrophoresis compared to wt or other transgenic lines.

LITERATURE CITED

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Bibliography Type: Description: (Last Updated:)